

Human-Machine Interface through Electromyography



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Electromyography (EMG) is a technique of recording muscular activities by measuring the potential difference across a muscle of interest. EMG signal acquisition is the latest trend as it paves way for a new method of human-machine interface (HMI). Here, the signals sent by brain are received as a potential difference across the muscle. The signals are very weak in strength and are inherently embedded in noise and a large offset of the electrodes; therefore special procedures are often required to extract EMG signals of good quality.

The circuit described here helps in making the analogue front end (AFE) of a simple and cost-effective EMG acquisition system. The circuit is specially designed for battery-operated portable systems. You can see the

acquired EMG signals on a PC-based oscilloscope or a normal oscilloscope for simplicity. One such PC-based oscilloscope (freeware) can be downloaded from http://www.zeitnitz.eu/scope/scope_141.exe

Circuit and working

The circuit diagram of HMI through EMG is shown in Fig. 3. Starting from the left, two electrodes attached to a muscle are connected to a single-supply instrumentation amplifier INA122 (IC1), which amplifies the voltage difference between the two electrodes, rejecting any signal that is common to both—thereby removing much of the noise and 50Hz AC EMI. Resistor R1 determines the gain of the output of IC1.

It is difficult to deal with the bi-polar nature of the bio-signals. To overcome this problem, we use virtual-ground topology or split-rail topology, which splits the main power rail into two rails—one having V_{cc} and the other with half of V_{cc} (virtual ground). The created virtual ground acts as the pseudo-zero voltage and thus makes the output to swing between 0 and V_{cc} with virtual ground as zero volts, which makes the entire portion of the EMG wave available and also in the positive range only, thus making this topology ideal for interfacing with digital systems

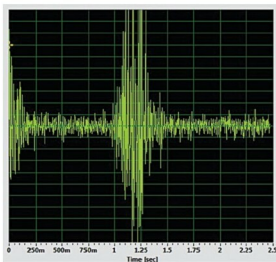


Fig. 1: EMG signal on the PC-based oscilloscope

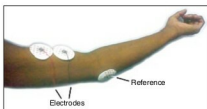


Fig. 2: Electrode placement

Test Points

Test point	Details
TP0	0V, GND
TP1	9V
Across TP2 and TP3	Around 500mV when muscles are not active
Across TP2 and TP3	Around 5V-6V when muscles are active

PARTS LIST

Semiconductors:

- IC1 - INA122 op-amp
- IC2 - OPA2241 op-amp

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-kilo-ohm
- R2, R3 - 10-kilo-ohm
- R4 - 1-kilo-ohm
- VR1 - 100-kilo-ohm preset

Capacitors:

- C1, C2 - 1 μ F, 25V electrolytic

Miscellaneous:

- BATT.1 - 9V battery
- S1 - On/off switch
- CON1 - 3-pin connector
- CON2 - 2-pin connector
- ECG electrodes
- Ultrasound gel

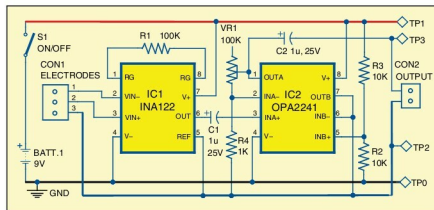


Fig. 3: Circuit diagram of HMI through EMG